

Philmont Scout Ranch Geology and Trek 12–15 Field Interpretation

Executive summary

Philmont is geologically richer than most backpackers realize. Trek 12–15 crosses a remarkably instructive transect from the eastern sedimentary benchlands of the Raton Basin, through Upper Cretaceous and Paleocene strata along the Cimarron and Ponil drainages, into the structurally uplifted and mineralized crystalline core around Baldy Mountain, and then back east into the coal-bearing, dinosaur-track-bearing, and archaeology-rich North Ponil country. In practical field terms, that means your trek moves from soft yellow-to-pink sedimentary slopes and caprock benches, into cliff-forming dacite porphyry sheets and mineralized fracture zones, then onto ancient Precambrian crystalline rocks, and finally back across canyon-country carved into Raton Basin strata. The official itinerary confirms the route sequence, major camps, daily mileage, and key program stops at Harlan, Devils Wash Basin, Cimarron River, Santa Claus, Baldy Town, Pueblano, Elkhorn, Ponil, Indian Writings, and the T-Rex track on the hike out. [filecite turn0file0](#)

The best primary geologic frame for this trek is still the combination of USGS Miscellaneous Geologic Investigations Map I-425 and USGS Professional Paper 505, supplemented by Robert Balk's trail guide and later regional stratigraphic work from the New Mexico Bureau of Geology and Mineral Resources and peer-reviewed papers. Those older Philmont-specific works remain very valuable because they were written for field observation at walking speed, not just for desk geologists. Where they need updating is mostly in nomenclature and precision: later work subdivides some units that PP 505 and I-425 lumped, especially the Triassic and Lower Cretaceous section; modern geochronology also refines the timing and context of regional volcanism in the Raton-Clayton volcanic field. [1](#)

For Trek 12–15 specifically, the highest-yield educational stops are these: the closed depressions and bench surfaces around Deer Lake Mesa and Devils Wash Basin; the Upper Cretaceous to Paleocene section around Ute Gulch, the Cimarron River, Ponil, and North Ponil; the dacite sheets, structural relief, and contact-zone geomorphology near Head of Dean and Wilson Mesa; the mineralized and mined terrain at Baldy Town, Black Horse, and French Henry; the petroglyph-bearing sandstone outcrops at Indian Writings; and the Maastrichtian dinosaur-track locality in North Ponil Canyon near the end of the trek. [filecite turn0file0](#)

[2](#)

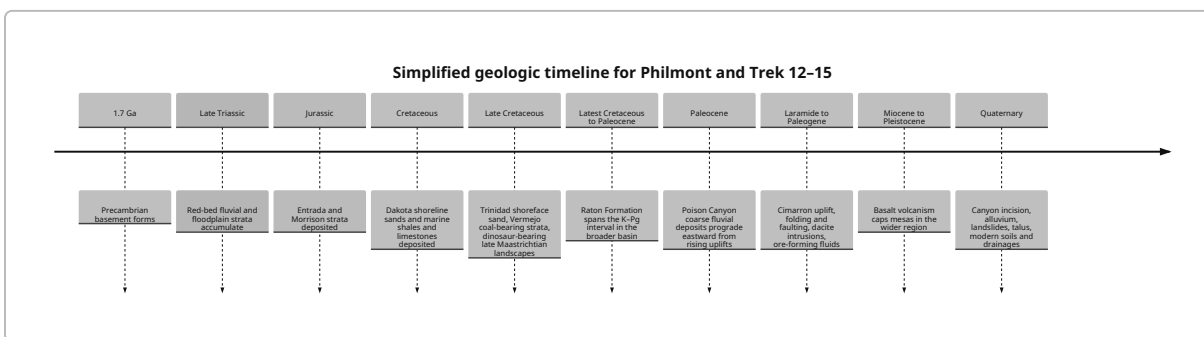
The single most important caution in using this report is spatial precision. Your uploaded itinerary gives camps, distances, gain/loss, and program waypoints, but it does **not** provide a georeferenced trail polyline or exact GPS waypoints. The day-by-day formation sequences below are therefore a **best-fit reconstruction** from the itinerary, the official camp sequence, and published geologic maps. They are appropriate for field interpretation, but not for centimeter-accurate GIS. [filecite turn0file0](#)

Regional geologic framework

Philmont sits at the eastern edge of the Cimarron Range, itself an eastern outlier of the southern Rocky Mountain system and a Laramide uplift bordering the Raton Basin. In regional terms, the trek traverses the boundary zone between a basement-cored mountain uplift and an adjacent sediment-filled basin. That is why you can hike from ancient crystalline rocks near Baldy into coal-bearing and dinosaur-bearing Cretaceous–Paleocene strata a day or two later. The range-basin relationship is the master control on the landscape, drainage, and relief you will see on Trek 12–15. ³

At Philmont, the oldest rocks exposed are Precambrian crystalline rocks, including granodioritic and metamorphic units, which form the resistant mountain core in the Baldy–Clear Creek area. Younger Mesozoic and Cenozoic sedimentary rocks flank that core to the east and northeast. These include Jurassic and Cretaceous clastic and marine rocks, followed by the coal-bearing Vermejo and Raton Formations, and then the coarse fluvial Poison Canyon Formation. Later igneous intrusions, including dacite porphyry sheets and related bodies, cut the sedimentary pile and locally helped create the dramatic ledges and ridges so characteristic of central and northern Philmont. Younger volcanic rocks, especially basalt flows related to the broader Raton–Clayton volcanic field, cap some mesas and benchlands. ⁴

Modern regional volcanologic work places Philmont’s younger mafic volcanism within the Raton–Clayton volcanic field, a Miocene-to-Pleistocene volcanic province tied to the northeastern end of the Jemez Lineament and influenced by extensional tectonics associated with the Rio Grande rift system. That broader regional context was not available to PP 505 in its current level of detail. What has **not** changed is the field lesson: harder volcanic rocks and intrusive bodies commonly make the cliffs, caps, and ribs, while softer shales and finer clastics tend to weather into slopes, swales, and grassy valleys. ⁵



This timeline is a trek-scale synthesis of the stratigraphic and tectonic framework used in PP 505, USGS I-425, later northeastern New Mexico summaries, and modern volcanic-field interpretations. ⁶

Stratigraphic framework and field identification

The formations and rock packages below are the ones most relevant to Trek 12–15. Ages and field identifications are simplified for use on foot. Exact GPS points are not specified in the uploaded itinerary, so the “waypoint examples” column uses named places along your trek rather than surveyed coordinates. The unit descriptions synthesize the Philmont map/guide literature with later northeastern New Mexico stratigraphic refinements. ⁷

Formation or rock package	Approximate age	Lithology	Diagnostic field cues	Most likely Trek 12–15 occurrences	Waypoint examples
Quaternary alluvium and colluvium	Quaternary	Gravel, sand, silt, slope wash, talus	Loose stream gravel, floodplain flats, unstable slope aprons, fresh cutbanks	Major drainages and valley bottoms	Cimarron River, Ponil Creek, North Ponil, Ute Gulch
Basalt lava caps	Neogene to Quaternary in regional context	Dark gray to black basalt, vesicular in places	Dark caprock, resistant mesa surfaces, blocky float, locally reddish cindery material	Some mesa/bench cap areas on the broader route margins	Around mesa country near Deer Lake/benchlands; broader Philmont context
Dacite porphyry intrusive sheets and dikes	Paleogene	Light gray to tan porphyry with conspicuous feldspar phenocrysts in finer groundmass	Cliff-forming ledges, rib-like ridges, pale salt-and-pepper appearance at distance, baked country rock near contacts	Head of Dean, Wilson Mesa belt, central Philmont cliffs and ribs	Head of Dean, ridge belts west of Cimarron River, French Henry vicinity
Poison Canyon Formation	Latest Cretaceous to Paleocene	Coarse yellow sandstone, conglomerate, sandy shale	Broad benchlands, ledgy caps, coarse channel bodies, yellow-orange weathering	Harlan–Devils Wash Basin–Head of Dean–Elkhorn uplands; caps above Ponil country	Harlan, Deer Lake Mesa, Devils Wash Basin, Elkhorn, uplands above North Ponil

Formation or rock package	Approximate age	Lithology	Diagnostic field cues	Most likely Trek 12–15 occurrences	Waypoint examples
Raton Formation	Latest Cretaceous to Paleocene	Sandstone, shale, coal, basal conglomerate	Pink to buff weathering, interbedded sandstone and shale, local mudcracks and cross-bedding, coal-bearing intervals regionally	North Ponil and Ponil country; much of the eastern/northeastern bench-and-canyon terrain	Ponil, Indian Writings approach, North Ponil country, lower-canyon benches
Vermejo Formation	Maastrichtian	Carbonaceous shale, soft friable sandstone, coal beds	Dark carbonaceous slopes, coal fragments, covered intervals between more resistant sandstone units	Local canyon sections below Raton, especially in coal-bearing country	Ponil Creek sections, North Ponil canyon exposures
Trinidad Sandstone	Late Cretaceous	Light gray quartzose shoreface sandstone	Persistent cliff or ledge, cleaner quartz sandstone than surrounding units	Local canyon walls and structural sections in the northern/eastern sedimentary country	Ute Gulch/ Cimarron River section, Ponil canyon sections

Formation or rock package	Approximate age	Lithology	Diagnostic field cues	Most likely Trek 12–15 occurrences	Waypoint examples
Lower and middle Upper Cretaceous units including Dakota and overlying marine section	Albian to Late Cretaceous	Quartz sandstone, marine shale, limestone	Dakota makes hard ledges; marine shales form softer slopes; limy horizons form thinner ledges	Best seen where canyons cut deeper into the section	Ute Gulch–upper Cimarron sector; some archaeology-bearing sandstone cliffs near Indian Writings may be in this broad sandstone package, though exact host needs on-site confirmation
Precambrian granodiorite, gneiss, schist	Proterozoic	Coarse crystalline felsic igneous and metamorphic rocks	Interlocking crystals, foliation in gneiss/schist, pinkish or gray coarse texture, sandy grus where weathered	Baldy massif and adjacent crystalline uplands	Baldy summit, Black Horse vicinity, west of Baldy Town

Compiled from Wanek et al. (1964), Miller et al. (1965), Balk (1953/2021 digital edition), and later northeastern New Mexico stratigraphic summaries and formation syntheses. ⁷

A few field-recognition rules will save you time on the trail. **Poison Canyon versus Raton** is the comparison you will make most often east of Baldy: Poison Canyon is usually the coarser, more ledge-forming, bench-capping unit, whereas Raton more commonly includes finer interbeds, coal/shale intervals, and pinker to buff slope-formers. **Dakota versus Trinidad** can be trickier, because both can look like resistant quartz sandstone; the discriminator is usually stratigraphic position and associated surrounding rocks rather than hand sample alone. **Dacite porphyry versus granodiorite** is another crucial pair: dacite porphyry shows larger crystals floating in a finer matrix, while granodiorite is coarse and crystalline all the way through. ⁸

Weathering and soils reinforce those differences. Shales and carbonaceous beds weather to gentler, grassier, and often muddier slopes; coarse sandstones and conglomerates weather to benches, cliffs, and rubbly ledges; dacite commonly breaks into angular blocks and talus; basalt creates dark, stony, caprock surfaces; and crystalline rocks around Baldy produce gritty sandy grus, blockfields, and resistant ridges. On

Trek 12–15, those weathering patterns will often tell you what you are standing on before the outcrop itself does. 9

Trek 12–15 day-by-day analysis

The official itinerary totals 66 miles. Daily distances and cumulative gain/loss below come directly from your uploaded itinerary. Geological interpretations are best-fit field reconstructions keyed to the camp sequence and named pass-through points. [filecite turn0file0](#)

Day	Segment	Miles	Gain	Loss
2	Turkey Creek Trailhead to Harlan	4.1	1,630 ft	710 ft
3	Harlan to Devils Wash Basin	6.1	2,490 ft	1,530 ft
4	Devils Wash Basin to Cimarron River	8.5	2,940 ft	4,170 ft
5	Cimarron River to Santa Claus	5.6	3,060 ft	1,870 ft
6	Santa Claus to Baldy Town	10.9	4,990 ft	3,580 ft
7	Baldy Town layover with Black Horse Mine and Baldy Mountain	6.3	4,050 ft	4,050 ft
8	Baldy Town to Pueblano via French Henry	6.6	2,570 ft	4,380 ft
9	Pueblano to Elkhorn	4.0	1,760 ft	1,310 ft
10	Elkhorn to Ponil	5.1	2,100 ft	3,490 ft
11	Ponil to Indian Writings via Hart Peak	4.5	1,910 ft	2,080 ft
12	Indian Writings to Camping HQ via T-Rex Track and Six Mile Gate	3.8	1,260 ft	1,500 ft

Source: uploaded itinerary. [filecite turn0file0](#)

Day 2: Turkey Creek Trailhead to Harlan.

This is your entry into the eastern and central benchlands. The most probable bedrock along this segment is the Poison Canyon bench country, with local alluvium in draws and scattered younger cover. What to watch for: broad orange-yellow sandstone slopes, coarse fluvial sandstone or conglomeratic float, and the strong topographic expression of resistant cap units over weaker underlying intervals. Educationally, this is the day to introduce the Raton Basin story: rivers draining a rising mountain belt spread coarse sediment eastward, building the very benches you are now climbing. A good photo or sketch here is a bench-and-slope profile showing how resistant beds define the skyline. Leave No Trace issue: do not shortcut on erodible sandy slopes, which degrade fast under concentrated foot traffic. [filecite turn0file0](#) 10

Day 3: Harlan to Devils Wash Basin.

This is one of the most geologically distinctive days of the trek because it carries you onto Deer Lake Mesa and the Devils Wash Basin area. PP 505 treated Deer Lake and Devils Wash as unusual closed depressions on the bench surface and even explored a wind-scour hypothesis; whether or not that mechanism is the full explanation, the field fact remains that this is an anomalous inward-draining mesa-top landscape in rocks otherwise associated with outward-draining benchlands. In the field, look for caprock-controlled

topography, shallow inward slopes, perched wet areas, and views that show the contrast between mesa-top surfaces and dissected canyons around them. Talking point: geology is not only rock type; it is also the geometry of surfaces and drainage. Suggested sketch: a plan-view sketch of Deer Lake/Devils Wash and the surrounding rim. 11

Day 4: Devils Wash Basin to Cimarron River via Ute Gulch and Cimarroncita.

This is likely your first really instructive sedimentary-section day. As you descend off the benchlands and cross Ute Gulch toward the Cimarron River, you should expect to pass from coarse bench-forming units into softer Upper Cretaceous marine shales and sandstones, with local dacite sheets entering the section near the mountain front. The upper Cimarron/Ute Gulch country is a classic place to see why some layers make cliffs and others make covered slopes. If exposures cooperate, look for gray to black shale slopes, harder quartzose sandstone ledges, and local intrusive ledges of dacite porphyry. Interpretive talking point: this valley compresses a long history—from shoreline sand to offshore mud to later igneous intrusion—into a short hiking distance. Safety note: shale slopes and landslide-prone covered intervals can be slick and unstable after rain.

Day 5: Cimarron River to Santa Claus.

This day likely climbs back out of the river section toward more bench-forming sedimentary terrain on the way to Santa Claus. Expect a transition from alluvium and lower canyon exposures back into higher-position sandstone and conglomeratic units, most likely dominated again by Poison Canyon country with local structural complexity. This is a good day to pay attention to springs and seeps: many occur where permeable sandstone overlies less permeable shale. That relation is one of the most practical geology lessons in Philmont because it predicts both vegetation and water. Suggested photos: one spring or seep line, one contact between a resistant ledge and covered slope, and one panorama back toward the Cimarron valley.

Day 6: Santa Claus to Baldy Town via Head of Dean.

This is the tectonic and lithologic transition day. You move from the eastern sedimentary benchlands into the central intrusive belt and then toward the Baldy crystalline core. Head of Dean is one of the best viewpoints on the whole itinerary because it lets you “read” Philmont: soft sedimentary benches eastward, more resistant intrusive ribs and cliffs nearer the mountain front, and the high crystalline mass of Baldy beyond. Outcrop clues should shift accordingly: more dacite porphyry ledges, more angular talus, less uniform bedded sandstone, and eventually more coarse crystalline float. The educational opportunity here is to teach the distinction between strata that were deposited and bodies that were intruded later. Safety note: this is a big mileage and elevation day; hydration, pacing, and storm awareness matter more than geology if weather turns.

Day 7: Baldy Town layover, Black Horse Mine, and Baldy Mountain.

This is the geological core day of the entire trek. Baldy Town and the summit day put you in the crystalline heart of the range and its mineralized fracture systems. Expect coarse crystalline rocks, local foliation in metamorphic rocks, quartz-rich veins, mine dumps, altered rock, and evidence of ore-forming fluids in fractures associated with igneous activity. PP 505 interprets the district broadly as a hot-fluid system related to granitelike intrusions and associated dacite bodies; that remains the correct field-scale model. What to look for safely, from legal and stable viewpoints only: quartz vein material, iron-oxide staining, pyrite/chalcopyrite-derived oxidation colors, and mine-waste textures. Interpretive talking point: miners were following structure as much as ore grade—fractures, contacts, and altered zones mattered. Safety/LNT note: do not enter adits, climb on waste piles, or remove artifacts. Photographs worth taking: one close-up

of crystalline texture, one mine-landscape overview, and one skyline view from Baldy showing the contrast between mountain core and basin fill.   14

Day 8: Baldy Town to Pueblano via French Henry.

This day descends from ore-country through one of the most important mining stops on the trek. French Henry is historically famous, but geologically it matters because it shows how the Baldy district mineralization persisted beyond the core camp area into additional workings along structural and lithologic opportunities. You are also transitioning back from crystalline/intrusive terrain into sedimentary canyon country. In the field, look for the change in debris type underfoot: coarse crystalline and altered mine waste should gradually give way to more bedded sedimentary float and softer slope materials lower down. Educational opportunity: compare hard-rock mining landscapes with the later logging landscape at Pueblano; both are human responses to geologic and topographic opportunity. Safety note: the mine setting is the place to repeat the strict no-entry, no-collecting rule.

Day 9: Pueblano to Elkhorn.

The geology takes a back seat to conservation programming on this short day, but the route itself is still instructive because it likely climbs from canyon-bottom Raton country back toward higher bench surfaces and interfluvies. In practical terms, this is a good day to focus on geomorphology rather than stratigraphy: Why is the canyon where it is? Why are some side slopes smooth and others rubbly? How do burn scars, soils, and runoff change what geology looks like? The itinerary's explicit warning about burn recovery, sun exposure, weather hazards, and early-hike timing is especially relevant here because post-fire terrain often strips vegetation from already-erodible slopes and makes rock/soil contrasts more obvious but also more fragile.

Day 10: Elkhorn to Ponil.

This is an excellent day for understanding North Ponil stratigraphy and topography. The route likely crosses upland caprocks and then descends into the structurally lower and more deeply incised Ponil country, where Raton and related units are especially important. Ponil is where you can really start to see the bench-and-canyon architecture of the northern ranch: coarser ledge-forming units higher, more coal-bearing and shaly intervals lower, and stream incision exploiting weaknesses in the section. Interpretive talking point: the North Ponil area is a landscape made by differential erosion, not just by uplift. Suggested photo set: wide canyon panorama, one cross-bedded sandstone exposure, and one slope-cleft contact between ledge and covered interval.

Day 11: Ponil to Indian Writings via Hart Peak.

This is where geology and archaeology come together most directly. The petroglyphs at Indian Writings occur on a sandstone cliff face in North Ponil country; on the ground, the exact stratigraphic host should be confirmed rather than assumed, but the important field lesson is clear: durable, exposed sandstone created a stable rock surface suitable for carving and preservation. Hart Peak adds the panorama piece—use it to look back across the benchlands and canyons and identify how lithology shapes the skyline. Educational opportunity: explain that archaeological site placement is often partly geologic—people use durable rock shelters, prominent walls, vantage points, and water-controlled corridors. LNT note: no touching, tracing, chalking, or rubbing of petroglyphs.

Day 12: Indian Writings to Camping HQ via the T-Rex Track and Six Mile Gate.

This final day is paleontologically outsized. The Philmont tyrannosaur track was discovered in North Ponil Canyon and formally described as *Tyrannosauripus pillmorei* close to the Cretaceous–Paleogene boundary.

Even if you do not see textbook-perfect exposures at every stop, the correct interpretive point is that this part of the trek walks through the same latest Cretaceous–earliest Paleogene basin story that makes the Raton Basin so important: coal, floodplains, channels, mudflats, the K–Pg interval, and now a globally famous track. What hikers should look for is not just the track itself, but the surrounding sedimentological evidence of a lowland wetland/fluvial environment: fine mudstone, sandstone bedding, possible mudcracks, carbonaceous material, and lateral changes in grain size. Safety and ethics: fossil localities are observation sites, not collecting sites. [filecite turn0file0](#) 18

For map work, the best overlay stack is straightforward: use the Philmont North and South sectionals as the hiking base; georeference your Trek 12–15 trail line if you obtain it later; drape simplified geologic polygons from USGS I-425 or a modernized derivative; then symbolize only six field classes for readability—Quaternary cover, basalt, dacite intrusions, Poison Canyon, Raton/Vermejo/Trinidad, and Precambrian crystalline rocks. That is enough detail to be useful on trail without overwhelming the page. Because your current itinerary does not include exact GPS points, this simplified overlay is the right level of precision for a field booklet. [filecite turn0file0](#)

Revisions and refinements since USGS Professional Paper 505

PP 505 remains excellent on field observation, landform explanation, and practical “how to read the landscape” geology. Still, several parts of the interpretive framework are now better constrained. The biggest update is **stratigraphic subdivision**. Older Philmont mapping and prose often grouped units more broadly than later workers prefer. In northeastern New Mexico, Triassic red beds once shown simply as Dockum are now commonly discussed in terms of formations such as the Baldy Hill, Travesser, and Sloan Canyon Formations, though nomenclature remains debated in places. Likewise, the Lower Cretaceous to lower Upper Cretaceous sandstone-and-shale interval once handled under broader labels such as Dakota is now interpreted with finer subdivisions and sequence-stratigraphic analysis, including units such as the Pajarito and Romeroville in regional work. 19

The second major update is **volcanic chronology and regional tectonic context**. PP 505 correctly recognized the importance of younger igneous rocks, but later work places Philmont’s basalt-capped mesas within a much more fully developed understanding of the Raton-Clayton volcanic field and its relationship to crustal weakness along the Jemez Lineament and broader extensional processes in New Mexico. For an advanced hiker, that matters because it explains why some mesas are lava-capped, why basalt and dacite need to be distinguished, and why northeastern New Mexico volcanism belongs to a regional story rather than a purely local one. 5

The third update is **paleontology**. PP 505 predates the famous Philmont tyrannosaur footprint by almost two decades, so one of the most compelling stops on Trek 12–15 simply did not exist in the published guide at the time. The North Ponil track substantially increases the interpretive value of the northern route because it ties Philmont directly to global discussions of latest Cretaceous terrestrial ecosystems and tyrannosaur ichnology. 20

A subtler refinement concerns **popular oversimplifications** of Philmont geology. Later nontechnical summaries often blur together the Baldy massif and the dacite porphyry ridges, as though the high country were one uniform intrusive body. The older USGS map and PP 505 were actually more careful than many later summaries: around Baldy, the terrain includes a genuine crystalline basement core as well as younger

intrusive sheets and dikes. In practice, hikers should not call every resistant light-colored outcrop “dacite” and move on. Baldy itself deserves a more discriminating eye. ²¹

Finally, some PP 505 geomorphic interpretations should be read as **field hypotheses**, not immutable doctrine. Deer Lake Mesa and Devils Wash Basin are a good example. PP 505’s discussion is thoughtful and still worth reading in the field, but a modern guide should present that explanation as one interpretive model among several, not as the last word. The practical lesson for your custom guide is simple: preserve the observational brilliance of PP 505, but flag the places where later mapping or modern geomorphic caution has narrowed certainty. ¹¹

Primary sources and prioritized bibliography

The most useful source stack for your custom guide is not huge. If you prioritize the references below, you will cover almost everything Trek 12–15 needs.

The single best **trek-specific primary source** remains the uploaded itinerary, because it fixes the actual camp sequence, day lengths, elevation changes, staffed-camp programs, and the explicit inclusion of Baldy, French Henry, Hart Peak, Indian Writings, and the T-Rex track. ¹²

The best **Philmont-scale geologic map** is:

- Wanek, A. A., Read, C. B., Robinson, G. D., Hayes, W. H., and McCallum, Malcolm. 1964. *Geologic map and sections of the Philmont Ranch region, New Mexico*. USGS Miscellaneous Geologic Investigations Map I-425.

Use this as the spatial backbone. It is still the most important single map for correlating camps, ridges, and formations.

The best **field-interpretive geologic text for hikers** is:

- Miller, J. P., Montgomery, A., Sutherland, P. K., and others. 1965. *Philmont Country: The Rocks and Landscape of a Famous New Mexico Ranch*. USGS Professional Paper 505.

Still the best prose guide for understanding what to look at on foot.

The best **trail-oriented Philmont geologic companion** is:

- Balk, Robert. 1953; digitized 2021. *Geologic Guide to the Trails at Philmont Scout Ranch*. New Mexico Bureau of Geology and Mineral Resources Open-File Report 22.

Especially valuable for turning geology into walking observations.

The most useful **modern regional update** is:

- Ziegler, K. E., Ramos, F. C., and Zimmerer, M. J. 2019. *Geology of Northeastern New Mexico, Union and Colfax Counties, New Mexico: A Geologic Summary*. New Mexico Geological Society Field Conference Series.

This is the cleanest bridge between older Philmont literature and modern regional stratigraphy and volcanology. ²²

For the **mountain-structure context**:

- **Smith, J. F., and Ray, L. L. 1943. *Geology of the Cimarron Range, New Mexico*. Geological Society of America Bulletin.**
Important for the uplift-and-basin framework. ²³

For the **coal-bearing basin stratigraphy**:

- **Pillmore, C. L. 1969. *Geology and coal deposits of the Raton coal field, Colfax County, New Mexico*.**
Useful for Vermejo, Raton, and basin evolution.
- **Lee, W. T. early 20th century USGS work on the Raton basin and Vermejo/Trinidad/Raton section.**
Still foundational for nomenclature behind the later maps. ²⁴

For the **mining district**:

- **Lindgren, W., Graton, L. C., and Gordon, C. H. 1910. *The Ore Deposits of New Mexico*. USGS.**
Still the classic ore-deposit reference for French Henry and the district.
- **McLemore, V. T. 2017. *Mining districts and prospect areas of New Mexico*. New Mexico Bureau of Geology and Mineral Resources Resource Map 24.**
Best concise modern district-level overview. ²⁵

For the **T-Rex track**:

- **Lockley, M. G., and Hunt, A. P. 1994. *A track of the giant theropod dinosaur Tyrannosaurus from close to the Cretaceous/Tertiary boundary, northern New Mexico*. Ichnos.**
This paper belongs in any serious Trek 12–15 guide because the day-12 stop is one of the trek's signature geologic moments. ²⁶

If you want to turn this report into a trail-ready booklet, the best next move is not more broad research. It is to build a **route-specific annotated map set** using the day sequence above and simplify the geology into field-readable classes. At Philmont, elegance matters more than exhaustiveness: one clean map, one paragraph of explanation, and one “what to look for” box per day will beat fifty extra pages every time.

¹ ⁵ ⁶ ²² https://en.wikipedia.org/wiki/Raton-Clayton_volcanic_field
https://en.wikipedia.org/wiki/Raton-Clayton_volcanic_field

² ¹⁴ https://en.wikipedia.org/wiki/Baldy_Mining_District
https://en.wikipedia.org/wiki/Baldy_Mining_District

³ ²³ https://en.wikipedia.org/wiki/Cimarron_Range%2C_New_Mexico
https://en.wikipedia.org/wiki/Cimarron_Range%2C_New_Mexico

⁴ ⁸ ⁹ ¹⁰ ¹¹ https://en.wikipedia.org/wiki/Poison_Canyon_Formation
https://en.wikipedia.org/wiki/Poison_Canyon_Formation

7 12 https://en.wikipedia.org/wiki/Glencairn_Formation

https://en.wikipedia.org/wiki/Glencairn_Formation

13 https://en.wikipedia.org/wiki/Tooth_of_Time

https://en.wikipedia.org/wiki/Tooth_of_Time

15 25 https://en.wikipedia.org/wiki/French_Henry_mine

https://en.wikipedia.org/wiki/French_Henry_mine

16 24 https://en.wikipedia.org/wiki/Raton_Formation

https://en.wikipedia.org/wiki/Raton_Formation

17 https://en.wikipedia.org/wiki/Philmont_Scout_Ranch

https://en.wikipedia.org/wiki/Philmont_Scout_Ranch

18 20 26 <https://en.wikipedia.org/wiki/Tyrannosauripus>

<https://en.wikipedia.org/wiki/Tyrannosauripus>

19 https://en.wikipedia.org/wiki/Pajarito_Formation

https://en.wikipedia.org/wiki/Pajarito_Formation

21 https://en.wikipedia.org/wiki/Geography_and_ecology_of_Philmont_Scout_Ranch

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